



| Degree                           | University/Institute                               | Year | CGPA/Marks(%) |
|----------------------------------|----------------------------------------------------|------|---------------|
| Ph.D (Mechanical- Robotics)      | IIT Hyderabad                                      | 2026 | 8.50          |
| MTech (Mechanical-Thermal Engg.) | NIT Mizoram                                        | 2022 | 8.80          |
| BE (Mechanical Engg.)            | Chhattisgarh Swami Vivekanand Technical University | 2019 | 81%           |
| XII (CGBSE- Maths)               | Saraswati Shiksha Mandir H S School Raipur         | 2015 | 89.00%        |
| X (CGBSE)                        | Saraswati Shiksha Mandir H S School Raipur         | 2013 | 89.50%        |

## SKILLS

**Robotics & Control:** ROS2, Gazebo, PX4, Isaac Sim, MATLAB, Flight Control Systems

**Computer Vision & AI:** OpenCV, PyTorch, TensorFlow, Machine Learning

**Programming:** Python, C++, MATLAB, Android App Development

**Hardware & IoT:** UAVs, Pixhawk, Jetson Nano/Orin, Raspberry Pi, LiDAR, IMUs, Embedded Systems.

**Manufacturing & Prototyping:** 3D Printing, CNC Machine Programming

**Design & Simulation:** ANSYS, SolidWorks, Creo Parametric, AutoCAD

## PROJECTS

- **Design and Implementation of Collaborative Payload Transport Using Four Quadcopters (DRDO Project)**
  - Developed a PX4-ROS2 based leader-follower UAV swarm with real-time RTK and MQTT communication, achieving successful collaborative outdoor transport of a 1 kg payload.
- **Passenger Drone (16 Rotor) – Ministry of Electronics and Information Technology (MEITY)**
  - Designed and developed a passenger-class drone; led structure, electronics, and control system design(75kg payload).
- **Tele-Echography System** – Acausal Automation, IIT Hyderabad
  - Developed a healthcare robotic platform using NEMA 27 motors, encoders, Dynamix steppers, and a Raspberry Pi.
- **Quadruped Robot** – IIT Hyderabad
- **BLDC Motor Cooling with PCM** – IIT Hyderabad, NIT Mizoram
  - Designed thermal management system using PCM for inner-rotor BLDC motor.
- **Prosthetic Hands (5 Types)** –SSIPMT Raipur
  - Developed flex-sensor, ultrasonic, force-sensor, mechanical, and gyro-controlled prosthetic prototypes.
- **3D Printer with IoT Control**
- **6-DOF Robotic Arm**
- **CNC Machines & Automation** – Multiple Prototypes
- Developed CNC drill/mill, wire-bending machine, CNC plotter, etc., integrating Arduino, CNC shields, stepper motors, and G-code programming.
- **IoT & Embedded Systems** – Multiple Prototypes
- Built an IoT-controlled boat, an IoT car, a home automation device, a self-charged wireless keyboard/mouse, and an automatic IoT gas stove (published in IJTRE).
- **Robotics & Smart Devices** – Multiple Prototypes
- Designed self-balancing robot, line follower, obstacle detector, surface detector, follower robot, smart blind stick, and smart dustbin.

## PUBLICATIONS / CONFERENCE PAPERS

1. **A. Sahu**, B. V. Harshavardhan, R. P. Kumar, “Autonomous Aerial Deployment of a Hybrid Drone–Quadruped System for Environmental Tasks”, Proceedings of the Institution of Mechanical Engineers, **Part C: Journal of Mechanical Engineering Science**, **SAGE Publications**.
2. **A. Sahu**, R. P. Kumar “Integrated Guidance and Control of a Mother–Child UAV–UGV System for Long-Endurance Cooperative Missions” in **Proceedings of the Journal of Field Robotics** (Under review).
3. **A. Sahu**, R. P. Kumar, H. Lalhmingsanga, “Experimental Validation of GO-PCM Thermal Management for Inner-Rotor BLDC Motors”, in Proceedings of the **Journal of power and energy** (Under review).
4. **A. Sahu**, N. Verma, “Automatic Gas Stove Control by Servo Motor and Wi-Fi”, **International Journal for Technological Research in Engineering (IJIRT)**, 2018.
5. **A. Sahu**, R. P. Kumar, “[Design and Implementation of Hexacopter Drone with Integrated Suction and Lift Mechanism with Real-Time Depth Sensing for Precision Object Handling](#)”, **ICRAE 2024**.
6. **A. Sahu**, R. P. Kumar, “[Design and Development of an Arm-Leg Hybrid Drone for Enhanced Aerial Manipulation and Mobility](#)”, **ROBIO 2024**.
7. N. S. Sridhar, **A. Sahu**, S. R. Reddy, R. P. Kumar, “[Collaborative Payload Transport Using a Four-Quadcopter Swarm with RTK and MQTT-Based Coordination](#)”, in Proceedings of the IEEE International Conference on Robotics and Mechatronics (**ICRM 2025**) .
8. **A. Sahu**, R. P. Kumar “ [Modeling and Control of a Two-Agent Aerial System for Dynamic Throw-and-Catch Maneuvers](#)”, in Proceedings of the IEEE International Conference on **APSCON 2026**.
9. **A. Sahu**, R. S. Ashtkar, K. S. V. Sreedath, L. K. Vemulapalli, R. P. Kumar, “Collaborative Outdoor Terrain Mapping via Real-Time Multi-Robot SLAM”, in Proceedings of the 7th International and 22nd National Conference on Machines and Mechanisms (**iNaCoMM 2025**).

10. **A. Sahu**, R. P. Kumar, “Risk-Aware Cooperative Aerial–Ground System for Adaptive Autonomous Navigation and Transport”, in Proceedings of the IEEE International Conference on **iConnect 2026**(Accepted)..
11. **A. Sahu**, R. P. Kumar " In-Flight Deployment and Vision-Guided Recovery of a Micro-UAV on an Autonomous Carrier Quadrotor ", in Proceedings of the IEEE International Conference on **IROS 2026**(Under review).
12. **A. Sahu**, R. Prasanth Kumar, “A Dual-Controller Auxiliary-Thruster Architecture for Wind Resilient multirotor Position Stabilization,” **INTSCon 2026** (Accepted).
13. V. V. S. Viswa Kiran, **A. Sahu** and R. Prasanth Kumar, “Vision–Language Guided Aerial Manipulation for Autonomous UAV Grasping,” Conference Paper. **CASE 2026** (Accepted).

#### PATENTS

1. 202541045613 - An Autonomous Position Stabilization System for Multirotor (**Granted**)
2. 202541045607 - A Multi-Level Unmanned Vehicle Deployment Platform(**Published**)
3. 202541044171 - An Under-Actuated Robotic Arm(**Published**)
4. 202541092644 - An Autonomous Multi-Arm Agriculture Drone (**Published**)
5. 202541063642 - Drone with Deployable Water Landing Platform for Multi-Environment Operations(**Published**)
6. 411555-001 – Hybrid Gripper Leg (**Granted**)
7. 411556-001 – Drone with Quadraped (**Granted**)
8. 411558-001 – Paddy Dryer Machine (**Granted**)
9. 411559-001 – Modular Robotic Manipulator Arm (**Granted**)
10. 412907-001 – Dual Arm Motor Connector (**Granted**)
11. 412910-001 – Tri Arm Connector (**Granted**)
12. 418897-001 – Forearm Connector (**Granted**)
13. 418898-001 – Motor Mounting Flange (**Granted**)
14. 411763-001 – Payload Locker Lever (**Granted**)
15. 444674-001 – Inclined Flanged Connector (**Granted**)
16. 411554-001 – Cooling System for BLDC Motor (**Granted**)
17. 411557-001 – Foldable Quadraped (**Granted**)
18. 453523-001 – Hexacopter Drone with Integrated Scissor Lift and Suction-Based Manipulation System (**Granted**)
19. 453524-001 – Multi-Level Drone Deployment Platform (**Granted**)
20. 453526-001 – Robotic Arm with Multi-Tool End Effector (**Granted**)
21. 453527-001 – Underactuated 4-DOF Robotic Arm (**Granted**)
22. 453520-001 – Agri-Drone with Multi-Tool Robotic Arm (**Granted**)
23. 453521-001 – Retractable and Foldable Dual-Motor Water Landing Gear for Drones (**Granted**)
24. 412909-001 – Landing Gear Connector (**Filed/Under Process**)
25. 419174-001 – Drone Cabin (**Filed/Under Process**)
26. 453519-001 – Adaptive Stabilization Drone with Secondary Propulsion System (**Filed/Under Process**)
27. 453525-001 – Multi-Tool End Effector (**Filed/Under Process**)
28. 453522-001 – Drone with Hybrid Arm-Leg Mechanism (**Filed/Under Process**)
29. 412908-001 – Hub Member with Inclined Flanges (**Filed/Under Process**)

#### INDUSTRIAL EXPERIENCE

1. **R&D Intern, Acausal Automation Pvt. Ltd., IIT Hyderabad** (Aug 2021 – Aug 2022)- Designed and manufactured innovative healthcare products from concept to prototype.
2. **Technical Executive, SAI Kripa Construction, Chhattisgarh** (Dec 2019 – Jun 2020)- Managed technical and operational projects across Chhattisgarh.
3. **Technical Executive, Innovation Center for Applied Nanotechnology, Chhattisgarh** (Jun 2019 – Nov 2019)- Oversaw technical and operational aspects of nano-technology retail dispensers.

#### INTERNSHIP AND TRAINING

1. One-year paid Internship, **IIT Hyderabad** (2021–2022).
2. INSPIRE Internship Camp, **DST, New Delhi** (2013).
3. Summer Training in **Hydraulic Systems**, Success Gate, Laxmi Hydraulic Services, Raipur (2017).
4. Summer Training in **Molding, Casting & Manufacturing**, Prakash Industries Ltd., Raipur (2018).
5. **Mechanical Industrial Skill Enhancement Program**, Michelin India Pvt. Ltd., Tamil Nadu (Winter 2018).
6. **Vocational Training**, Bhilai Steel Plant (SAIL), Chhattisgarh – *Excellent Performance* (2017).
7. **Bloom & Billet Training**, Prakash Industries Ltd., Raipur (2016).
8. **CAD/CAM (AutoCAD)** – 100 hrs, CADD Centre, Raipur (2016).
9. **CAD/CAM (Creo)** – 300 hrs, PTC University,SSIPMT Raipur (2017).
10. **CAD/CAM (SolidWorks)** – 300 hrs, SSIPMT Raipur (2018).

#### WORKSHOP

1. Organized “**Drone Technology for Urban Mobility**” Workshop, IIT Hyderabad (Mar 2023).
2. Organized **Structural Integrity Conference & Exhibition (SICE)**, IIT Hyderabad (2022).
3. **Automotive CI Engine** Workshop – 15 days, SSIPMT Raipur (2019).
4. **Embedded Systems** Workshop – 15 days, SSIPMT Raipur (2018).
5. **Quadcopter Drone** Workshop – 7 days, SSIPMT Raipur (2015).

#### ACHIEVEMENTS AND AWARDS

- Selected as **Junior Research Fellow (JRF)**, IIT Bhilai (2021).
- Awarded **Paid Internship**, IIT Hyderabad (2021).
- **National Chhatra Vishwakarma Award 2019** – 3rd Rank, felicitated by Hon. Vice President of India.
- Recipient of **National Scholarship**, Government of India (B.E.).

- Recipient of **Merit Scholarship**, Government of Chhattisgarh (B.E.).
- Winner – **Mobile Controlled Robotics Competition**,SSIPMT (Times World Group).
- 1st Rank – **District-level Science & Engineering Model Competition**.
- 1st Rank – **District-level Science Competition**.
- 1st Rank – **District-level Bhartiya Sanskriti Gyan Competition**.
- 2nd Rank – **University-level Model Competition**.
- 1st Rank – **College-level Robotics Competition**.
- Awarded **INSPIRE Scholarship**, Department of Science & Technology, Govt. of India (2015).

#### CERTIFICATES

- **Techkriti 2018** – All India Research Championship on Artificial Intelligence, IIT Kanpur @ NIT Raipur.
- **Publication Certificate** – *International Journal for Technological Research in Engineering*.
- **Campus Ambassador**, SSIPMT – All India Council for Robotics & Automation (AICRA), Delhi.
- **Diploma in AutoCAD**, CADD Centre, Raipur.
- **Certified in CREO Parametric**, PTC University.
- Workshop Certificate – **Automotive CI Engine**.